

Power equations



1

a $x = 3$

b $x = 8$

c $x = \frac{8}{5}$

d $x = 6$

e $x = 2$

f $x = 6$

g $x = 5$

h $x = 2$

Exponential equations

2

a $x = 8$

b $x = -4$

c $x = \frac{2}{9}$

d $x = 3$

e $x = 0$

f $x = -2$

g $x = 7$

h $x = -2$

i $x = -3$

j $x = \frac{1}{3}$

k $x = \frac{1}{2}$

l $n = \frac{1}{3}$

3

a $3 < x < 4$

b $3 < x < 4$

c $-4 < x < -3$

d There is no real solution.

4

a i $2^{3x} = 2^2$

ii $x = \frac{2}{3}$

b i $2^{4x} = 2^{-1}$

ii $x = -\frac{1}{4}$

c i $2^{-10} = 2^{2x}$

ii $x = -5$

d i $2^{\frac{x}{2}} = 2$

ii $x = 2$

5

a $y = \frac{3}{2}$

b $x = 6$

c $y = 4$

d $k = -2$

e $x = 2$

f $x = -2$

g $x = -3$

h $x = \frac{7}{3}$

i $x = -7$

j $x = \frac{21}{19}$

k $y = \frac{11}{6}$

l $x = -\frac{41}{6}$

m $x = 5$

n $x = 1$

o $x = -\frac{11}{2}$

p $x = \frac{21}{4}$

q $x = -7$

r $x = 2$

s $x = 5$

t $x = \frac{5}{2}$

u $x = 4, x = -1$

v $x = 3$

6

a i 3^{x+nx}

ii $x = \frac{4}{1+n}$

b i 3^{3x-2k}

ii $x = 1 + \frac{2k}{3}$



7

a

i $m = 2^x$

ii $x = 3, 0$

b

i $m = 2^x$

ii $x = 3, 2$

c

i $m = 2^x$

ii $x = -1, 3$

d

i $m = 4^x$

ii $x = 0, 3$

e

i $m = 2^x$

ii $x = 0, 2$

f

i $m = 3^x$

ii $x = 1, 2$

8 8

Applications

9 $x = -2$

10 $h = 2$

11

a $n = 8$

b $k = 4$

c $m = -2$

12

a $t = 5$ hours

b $t = 10$ hours

c $t = 12$ hours

13

a 440 Hz

b 262 Hz

c $n = 73$